

Supplementary Appendix

This appendix has been provided by the authors to give readers additional information about their work.

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**Supplementary Materials for
Abiraterone plus Prednisone in Metastatic, Castration-Sensitive Prostate Cancer**

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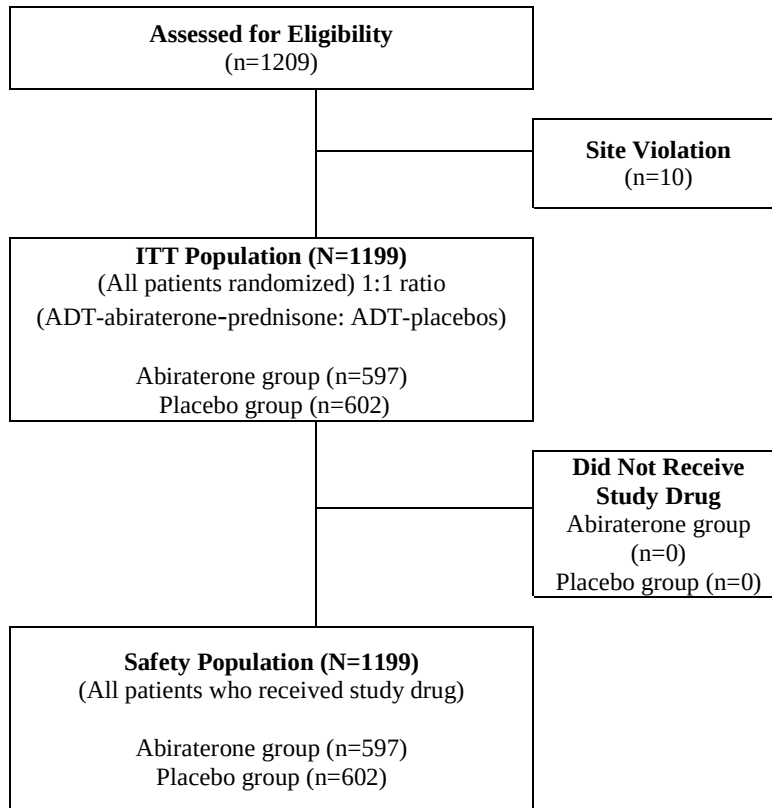
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List of Investigators for the LATITUDE Study

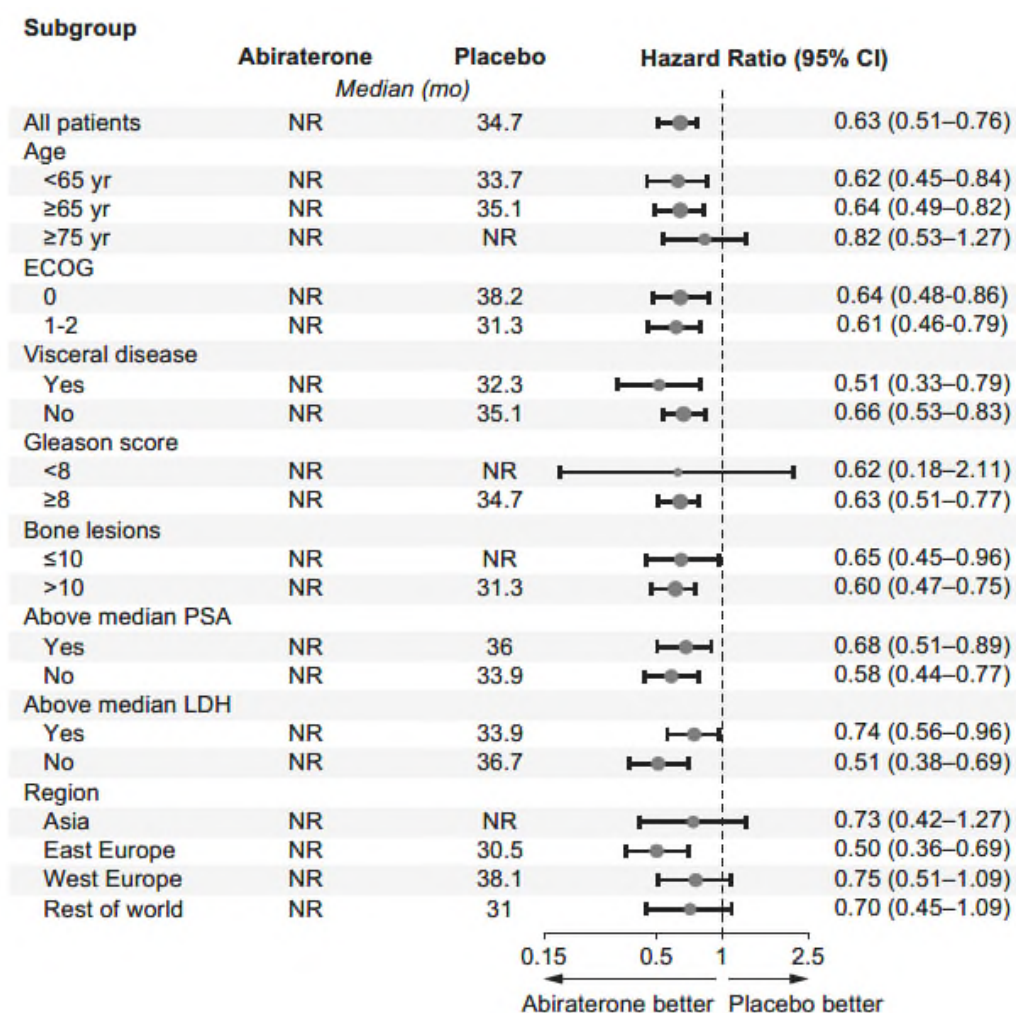
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Supplementary Figure S1. CONSORT Diagram.

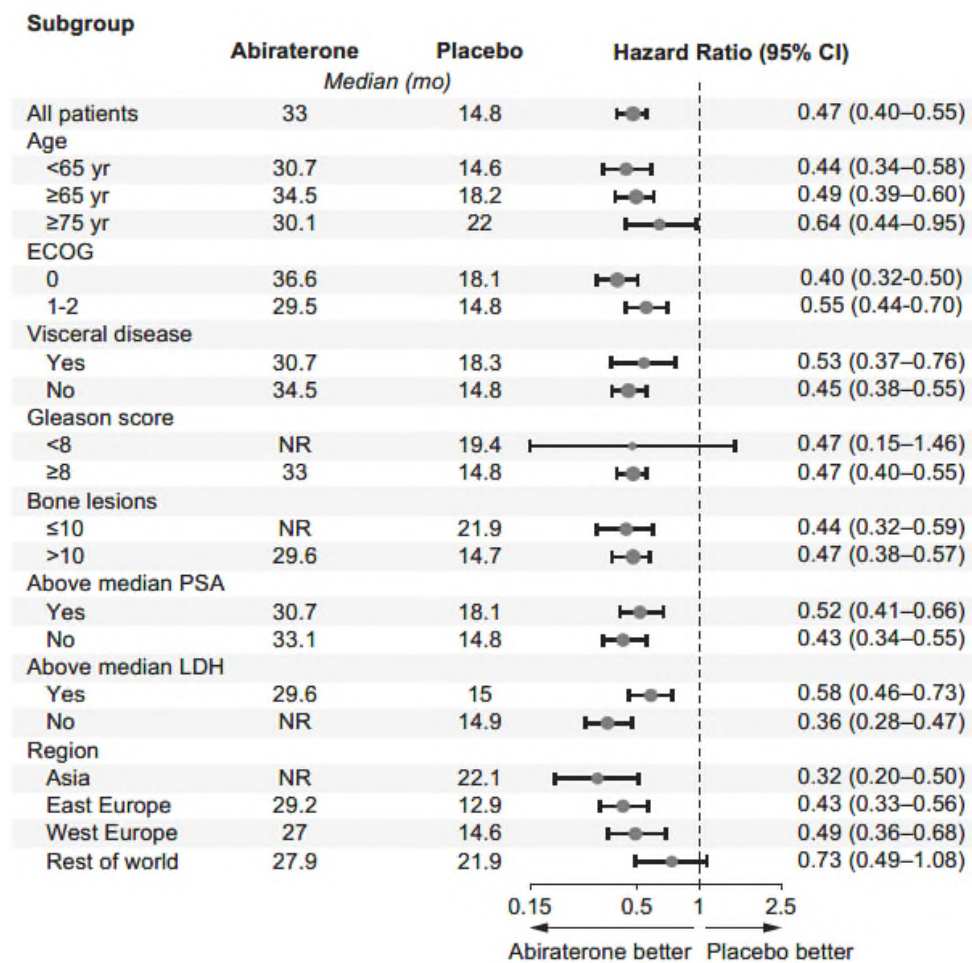


Supplementary Figure S2. Overall Survival Subgroup.

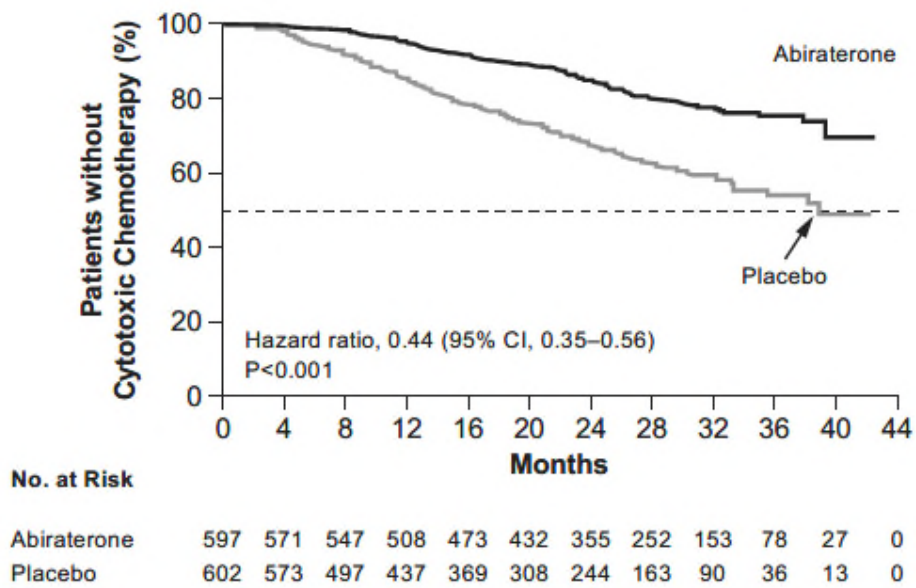


In Supplementary Figs. S2 and S3, the size of the circles reflects the number of patients affected and represents descriptive information, with nominal 95% confidence intervals. All analyses were performed with the use of a stratified log-rank test according to stratification factors. ECOG denotes Eastern Cooperative Oncology Group (ECOG) scale (a performance of 0 indicates asymptomatic, 1 restricted in strenuous activity but ambulatory, and 2 indicating ambulatory and up and about more than 50% of waking hours and is capable of all self-care but unable to carry out any work activities). Patient stratification included ECOG of 0 or 1 versus 2, but the subgroup analysis was performed on the basis of 0 versus 1 or 2 as there were few patients with a score of 2. LDH denotes lactate dehydrogenase and PSA prostate-specific antigen.

Supplementary Figure S3. Radiographic Progression-free Survival Subgroup.

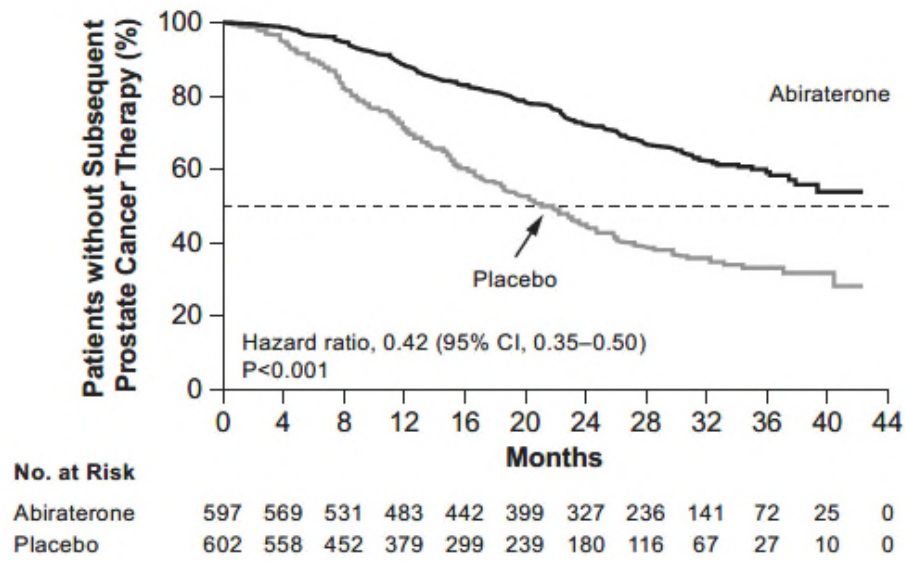


Supplementary Figure S4. Time to Initiation of Cytotoxic Chemotherapy.



The median was not reached in the abiraterone group and 38.9 months in the placebo group.

Supplementary Figure S5. Time to Subsequent Prostate Cancer Therapy



The median was not reached in the abiraterone group and 21.6 months in the placebo group.

Supplementary Table S1. Radiographic Progression-free Survival Definition per RECIST Version 1.1 as Adapted by Prostate Cancer Working Group 2 Criteria.

- Patients who had ≥ 2 new bone lesions on 16-week scans, with confirmatory scans performed 6 or more weeks later
 - o Patients with confirmatory scans that showed ≥ 2 new lesions compared with 16-week scans (i.e., total of four new lesions compared with baseline scan) were considered to have disease progression by bone scan
 - o Patients who on confirmatory scans did not show ≥ 2 new lesions compared with 16-week scans were not considered to have disease progression at that time. All subsequent scans served as “confirmatory scans” and were considered to show disease progression with ≥ 2 new lesions compared with 16-week scans
- Patients who did not have ≥ 2 new bone lesions on 16-week scans did not require a confirmatory scan. For any subsequent scan beyond 16 weeks, the first scan that would show ≥ 2 new lesions compared with baseline scans was considered to demonstrate disease progression by bone scan

RECIST denotes Response Evaluation Criteria In Solid Tumors.

Supplementary Table S2. Statistical Plan.

Overall Assumption	rPFS		OS
α	0.001		0.049
Power	94%		85%
HR	0.67		0.81
Expected events	565 (single analysis)	426, 554, 852 (two interim, one final analysis)	
Planned OS Analysis	Interim 1* (50% of total events)	Interim 2 (65% of total events)	Final
Projected observed OS events	~426	~554	~852
Efficacy boundary (HR)	0.78	0.81	0.87
Stopping P value under H ₀ (cumulative alpha spend)	(0.011)	(0.022)	(0.049)

*At time of final radiographic progression-free analysis. Hazard ratio (HR) denotes hazard ratio, H_0 no improvement, rPFS radiographic progression-free survival, and OS overall survival.

Two interim analyses were planned for overall survival in addition to the final analysis. The first analysis occurred after observing approximately 426 of the required total number of death events (50%), and the second interim analysis was planned after approximately 554 events (65%). A final analysis was planned for after the 852 events had occurred.

Supplementary Table S3. Demographics and Baseline Disease Characteristics.

	Abiraterone Group (n=597)	Placebo Group (n=602)
Age (yr), n (%)		
n	597	602
<65	221 (37)	233 (39)
65–69	112 (19)	134 (22)
70–74	141 (24)	115 (19)
≥75	123 (21)	120 (20)
Median	68.0	67.0
Range	38–89	33–92
Gleason score at initial diagnosis, n (%)		
n	597	602
<7	4 (0.7)	1 (0.2)
7	9 (2)	15 (2)
≥8	584 (98)	586 (97)
Baseline pain score (BPI-SF Item 3), n (%)		
n	570	579
0–1	284 (50)	288 (50)
2–3	123 (22)	137 (24)
≥4	163 (29)	154 (27)
Patients with ≥3 bone metastases at screening, n/N (%)	586/597 (98.2)	585/602 (97.2)
Patients with high risk at screening, n (%)		
n	597	601
Gleason score ≥8 + ≥3 bone lesions	573 (96)	569 (95)
Gleason score ≥8 + measurable visceral disease	82 (14)	87 (14)
≥3 bone lesions + measurable visceral disease	84 (14)	85 (14)
Gleason score ≥8 + ≥3 bone lesions + measurable visceral disease	71 (12)	70 (12)
Extent of disease, n (%)		
n	596	600
Bone	580 (97)	585 (98)
Liver	32 (5)	30 (5)
Lungs	73 (12)	72 (12)
Node	283 (47)	287 (48)
Prostate mass	151 (25)	154 (26)
Viscera	18 (3)	13 (2)
Soft tissue	9 (2)	15 (3)
Other	2 (0.3)	0
Patients with previous prostate cancer therapy, n (%)		
n	560	560

Radiotherapy [†]	19 (3)	26 (4)
Hormonal	501 (84)	501 (83)
GnRH agonists/antagonists	449 (75)	450 (75)
Orchiectomy	73 (12)	71 (12)
First-generation androgen receptor agonists	373 (62)	371 (62)
Other	7 (1)	10 (2)
Time from GnRH agonist/antagonist to first dose, (mo)		
n	445	449
Median	1.08	1.08
Range	0.1–3.0	0.1–3.5

Supplementary Table S4. Causes of Death.

	Abiraterone Group (n=597) n (%)	Placebo Group (n=602)
Causes of Death		
All deaths	169 (28)	237 (39)
Death due to prostate cancer	122 (20)	194 (32)
Adverse event	28 (5)	22 (4)
Other	10 (2)	11 (2)
Unknown	9 (2)	10 (2)
Adverse events leading to death	28 (5)	24 (4)
Cardiac disorders	10 (2)	6 (1)
Acute coronary syndrome	1 (0.2)	1 (0.2)
Acute myocardial infarction	1 (0.2)	0
Atrioventricular block complete	0	1 (0.2)
Cardiac arrest	2 (0.3)	1 (0.2)
Cardiac failure	1 (0.2)	0
Acute cardiac failure	0	1 (0.2)
Congestive cardiac failure	1 (0.2)	0
Cardio-respiratory arrest	2 (0.3)	2 (0.3)
Cardiopulmonary failure	1 (0.2)	0
Myocardial infarction	1 (0.2)	0
Gastric ulcer perforation	1 (0.2)	0
Intestinal ischemia	1 (0.2)	0
Intestinal obstruction	1 (0.2)	0

Multi-organ failure	1 (0.2)	2 (0.3)
Sudden death	2 (0.3)	4 (0.7)
Bronchopneumonia	0	1 (0.2)
Lower respiratory tract infection	1 (0.2)	0
Lung infection	1 (0.2)	0
Pneumonia	3 (0.5)	1 (0.2)
Sepsis	0	2 (0.3)
Urosepsis	0	1 (0.2)
Subdural hematoma	2 (0.3)	0
Traumatic intracranial hemorrhage	0	1 (0.2)
Arthralgia	0	1 (0.2)
Cerebral hemorrhage	1 (0.2)	0
Cerebrovascular accident	2 (0.3)	2 (0.3)
Coma	0	1 (0.2)
Optic nerve compression	0	1 (0.2)
Spinal cord compression	1 (0.2)	1 (0.2)
Suicide	0	1 (0.2)
Urinary retention	0	1 (0.2)
Aspiration	1 (0.2)	0
Pulmonary embolism	0	1 (0.2)
Respiratory failure	2 (0.3)	0

Supplementary Table S5. Subsequent Life-prolonging Therapy for Prostate Cancer.

Agent	Abiraterone Group (n=597)	Placebo Group (n=602)
	n (%)*	
	n=314	n=469
Abiraterone acetate plus prednisone	10 (3)	53 (11)
Cabazitaxel	11 (4)	30 (6)
Docetaxel	106 (34)	187 (40)
Enzalutamide	30 (10)	76 (16)
Radium-223	11 (4)	27 (6)

*Patients who discontinued treatment and were eligible for subsequent therapy. At the time of unblinding at the first interim OS analysis, patients who were still receiving drug in the double-blind treatment phase were allowed to crossover to ADT-abiraterone-prednisone. 793 patients remained alive and 369 patients were still receiving blinded study treatment (abiraterone group, n = 257; placebo group, n =112) and these 112 patients in the placebo group were eligible for crossover.

Supplementary Table S6. Adverse Events Leading to Treatment Discontinuation, Dose Reduction, and Dose Interruption.

Adverse Event	Abiraterone Group (n=597)	Placebo Group (n=602)
<i>no. of patients (%)</i> *		
Treatment Discontinuation		
All	73 (12)	61 (10)
Bone pain	3 (0.5)	6 (1)
Spinal cord compression	5 (0.8)	6 (1)
Dose Reduction		
All	55 (9)	17 (3)
ALT increased	23 (4)	5 (0.8)
AST increased	12 (2)	4 (0.7)
Hypertension	8 (1)	2 (0.3)
Hypokalemia	7 (1)	0
Dose Interruption		
All	183 (31)	102 (17)
ALT increased	30 (5)	10 (2)
AST increased	30 (5)	9 (2)
Hyperglycemia	6 (1)	4 (0.7)
Hypokalemia	47 (8)	4 (0.7)
Hypertension	39 (7)	15 (2)
Nausea	5 (0.8)	6 (1)

*Individual adverse events listed at $\geq 1\%$ in either group. Greater than 92% of patients in each treatment group had $>80\%$ compliance with study drug dosing.

ALT denotes alanine aminotransferase, and AST aspartate aminotransferase.

Supplementary Table S7. Ongoing Phase 3 Combination Therapy in Patients With Metastatic, Castration-naïve Prostate Cancer.

Study	Identifier	Groups	Enrolled Pts (N)	Primary End Point	Status
LATITUDE	NCT01715285	ADT ± AA/P	1209	rPFS, OS	Ongoing
STAMPEDE (Arm G)	NCT00268476	ADT ± AA/P	1800	OS	Ongoing
PEACE-1	NCT01957436	ADT ± DOC vs. ADT + A/P ± DOC (± local RT)	916	PFS, OS	Recruiting
STAMPEDE (Arm J)	NCT00268476	ADT ± AA/P + ENZ	1800	OS	Ongoing
SWOG-1216	NCT01809691	ADT + TAK-700 vs. ADT + BIC	1304	OS	Recruiting
ENZAMET	NCT02446405	ADT + ENZ vs. ADT + antiandrogen	1100	OS	Ongoing
TITAN	NCT02489318	ADT ± APA	1000	rPFS, OS	Recruiting
ARCHES	NCT02677896	ADT ± ENZ	1100	rPFS	Recruiting
ARASENS	NCT02799602	ADT + DOC ± ODM-201	1300	OS	Recruiting

*No prior cytotoxic chemotherapy, but up 2 cycles of docetaxel for metastatic disease is allowed.

AA denotes abiraterone acetate, ADT androgen deprivation therapy, APA apalutamide, BIC bicalutamide, DOC docetaxel, ENZ enzalutamide, OS overall survival, P prednisone, ODM-201 darolutamide, rPFS radiographic progression-free survival, RT radiotherapy, and TAK-700, orterenol.